TUGAS 02

Bab 2 — Konsep Container dan Instalasi Docker

Matakuliah Workshop DevOps

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Departemen Teknik Informatika dan Komputer

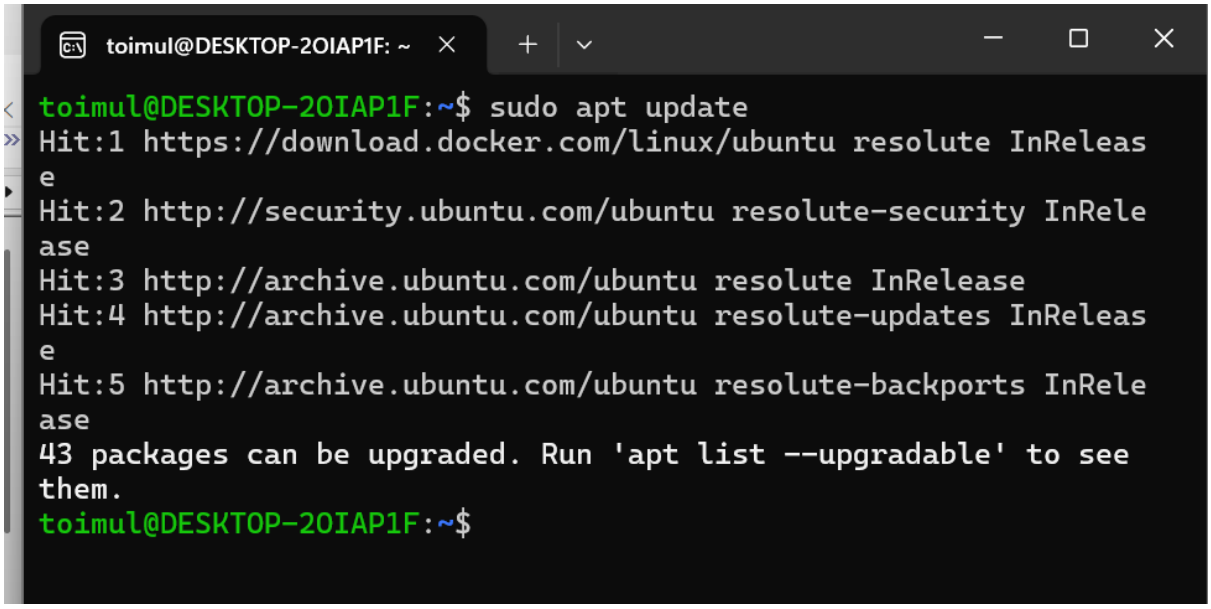
Program Studi Teknik Informatika

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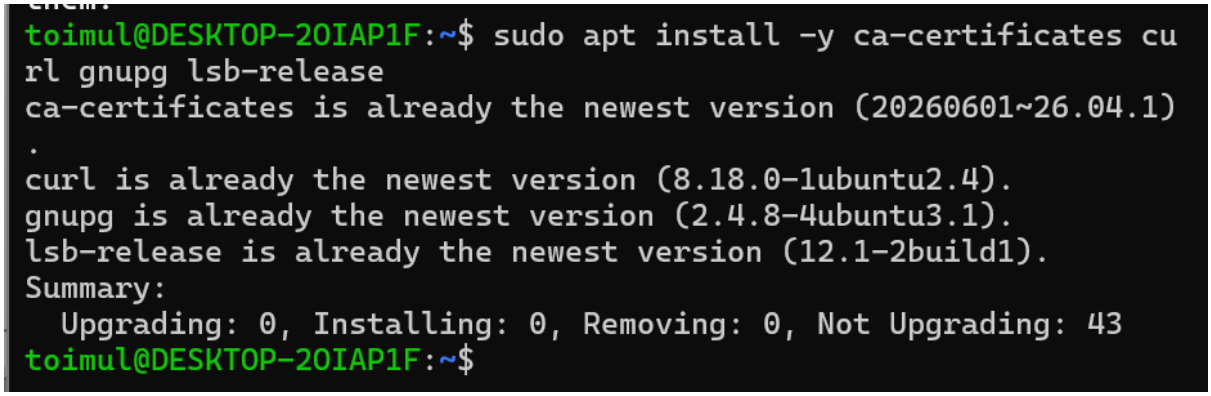
Langkah Praktikum Exploratif

1. Instalasi docker engine ubuntu

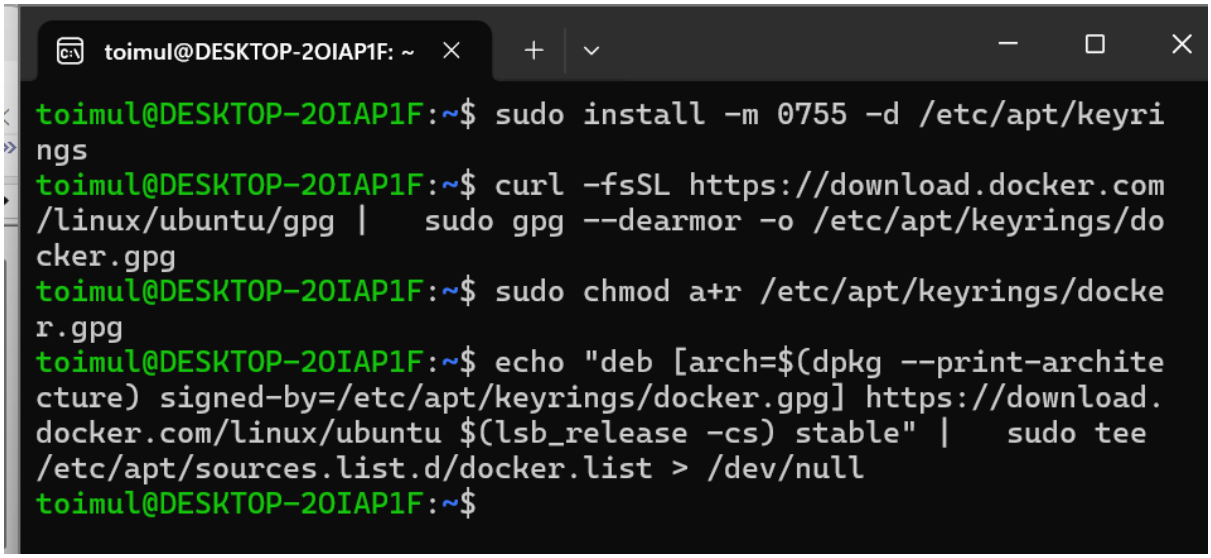
1.1. Pembaruan Repository Package



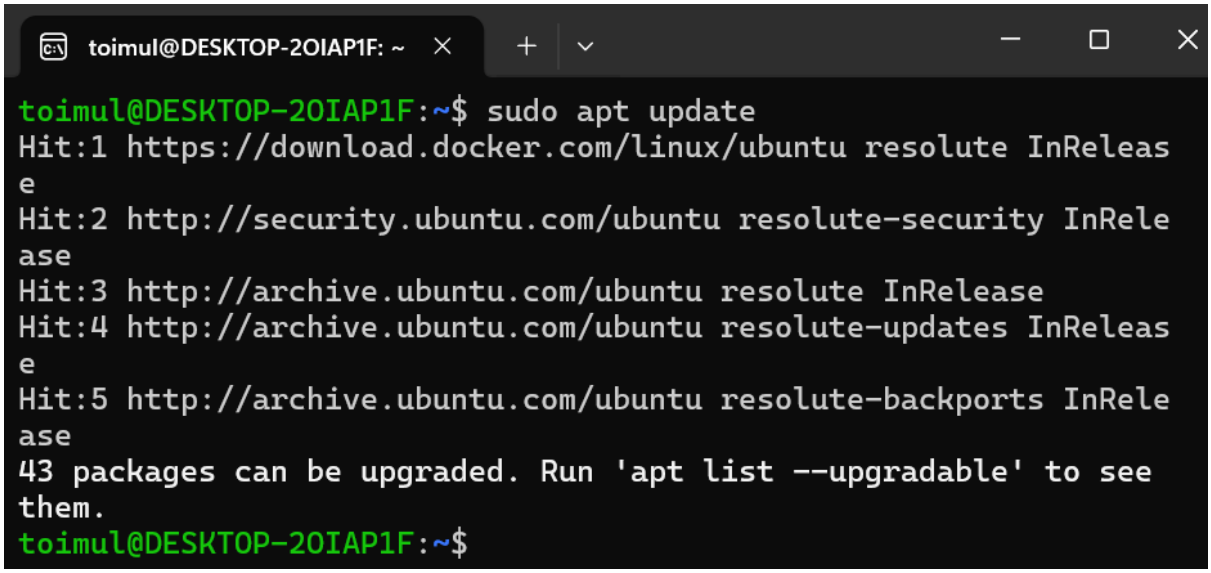
1.2. Instalasi Dependensi Dasar



1.3. Konfigurasi GPG Key & Repository Resmi Docker



1.4. Update Indeks Paket: untuk memuat paket dari repository Docker yang baru ditambahkan



1.5. Instalasi Docker Engine & Plugin

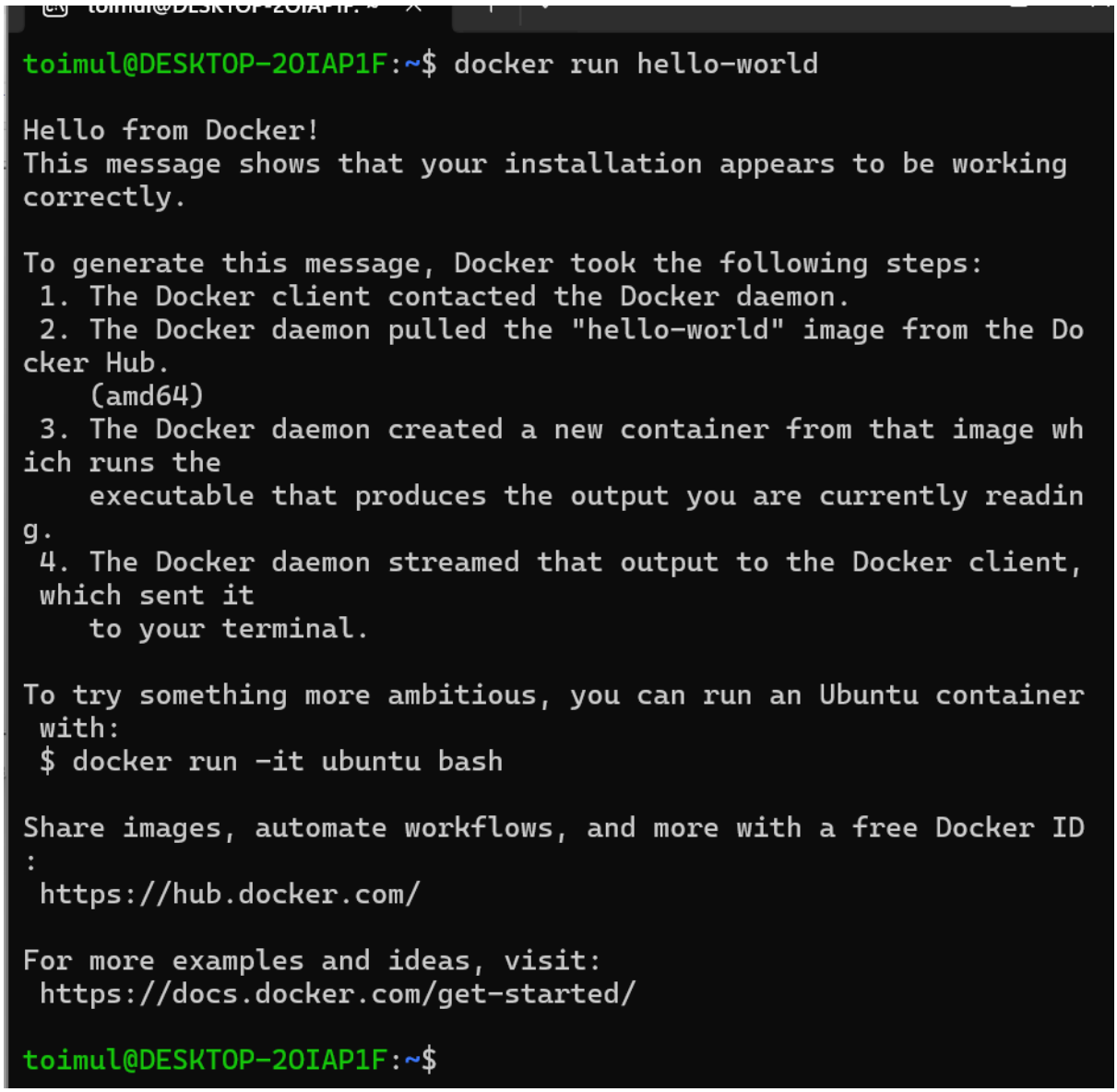
```
toimul@DESKTOP-2OIAP1F:~$ sudo apt install -y docker-ce docker-c
e-cli containerd.io docker-buildx-plugin docker-compose-plugin
docker-ce is already the newest version (5:29.8.0-1~ubuntu.26.04
~resolute).
docker-ce-cli is already the newest version (5:29.8.0-1~ubuntu.2
6.04~resolute).
containerd.io is already the newest version (2.3.4-2~ubuntu.26.0
4~resolute).
docker-buildx-plugin is already the newest version (0.37.0-1~ubu
ntu.26.04~resolute).
docker-compose-plugin is already the newest version (5.5.1-1~ubu
ntu.26.04~resolute).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 43
toimul@DESKTOP-2OIAP1F:~$
```

1.6. Konfigurasi Hak Akses User Non-Root

```
toimul@DESKTOP-2OIAP1F:~$ sudo usermod -aG docker $USER
toimul@DESKTOP-2OIAP1F:~$ newgrp docker
toimul@DESKTOP-2OIAP1F:~$ docker version
Client: Docker Engine - Community
 Version:           29.8.0
 API version:       1.56
 Go version:        go1.26.8
 Git commit:        88096ef
 Built:             Thu Sep  3 21:50:20 2026
 OS/Arch:           linux/amd64
 Context:           default

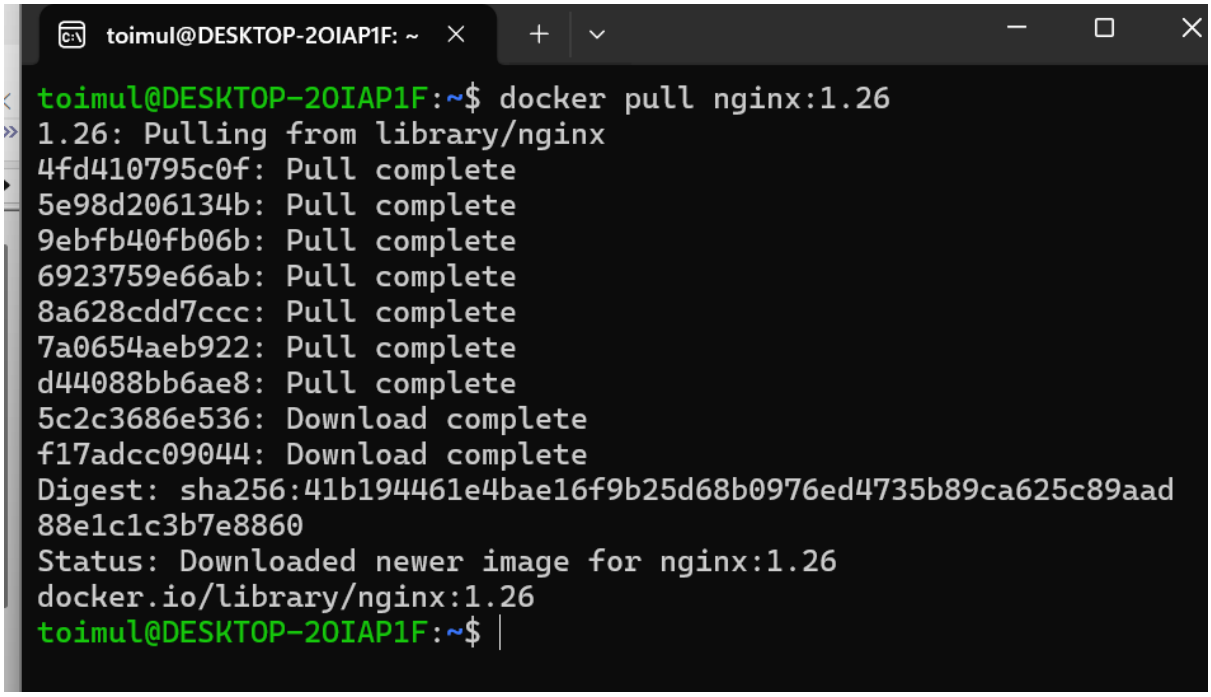
Server: Docker Engine - Community
 Engine:
  Version:          29.8.0
  API version:      1.56 (minimum version 1.40)
  Go version:       go1.26.8
  Git commit:       3ce5872
  Built:            Thu Sep  3 21:50:20 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.3.4
  GitCommit:        db8809540e1a7a9da5d518876894933ff55692ab
 runc:
  Version:          1.5.1
  GitCommit:        v1.5.1-0-g8f2685a4
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

1.7. Pengujian Container Pertama

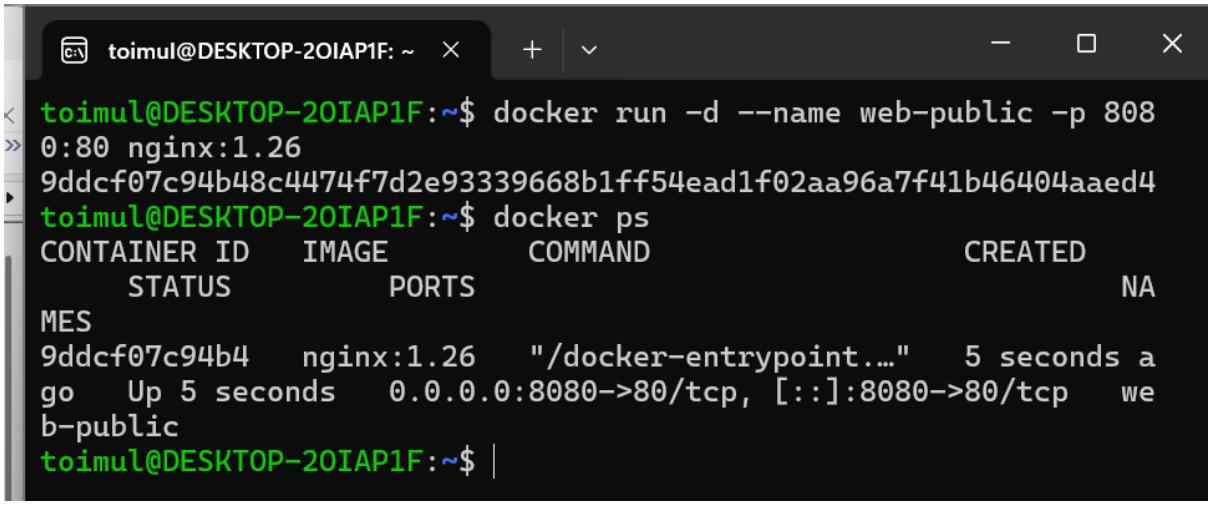


2. Container Nginx dan Ubuntu Interaktif

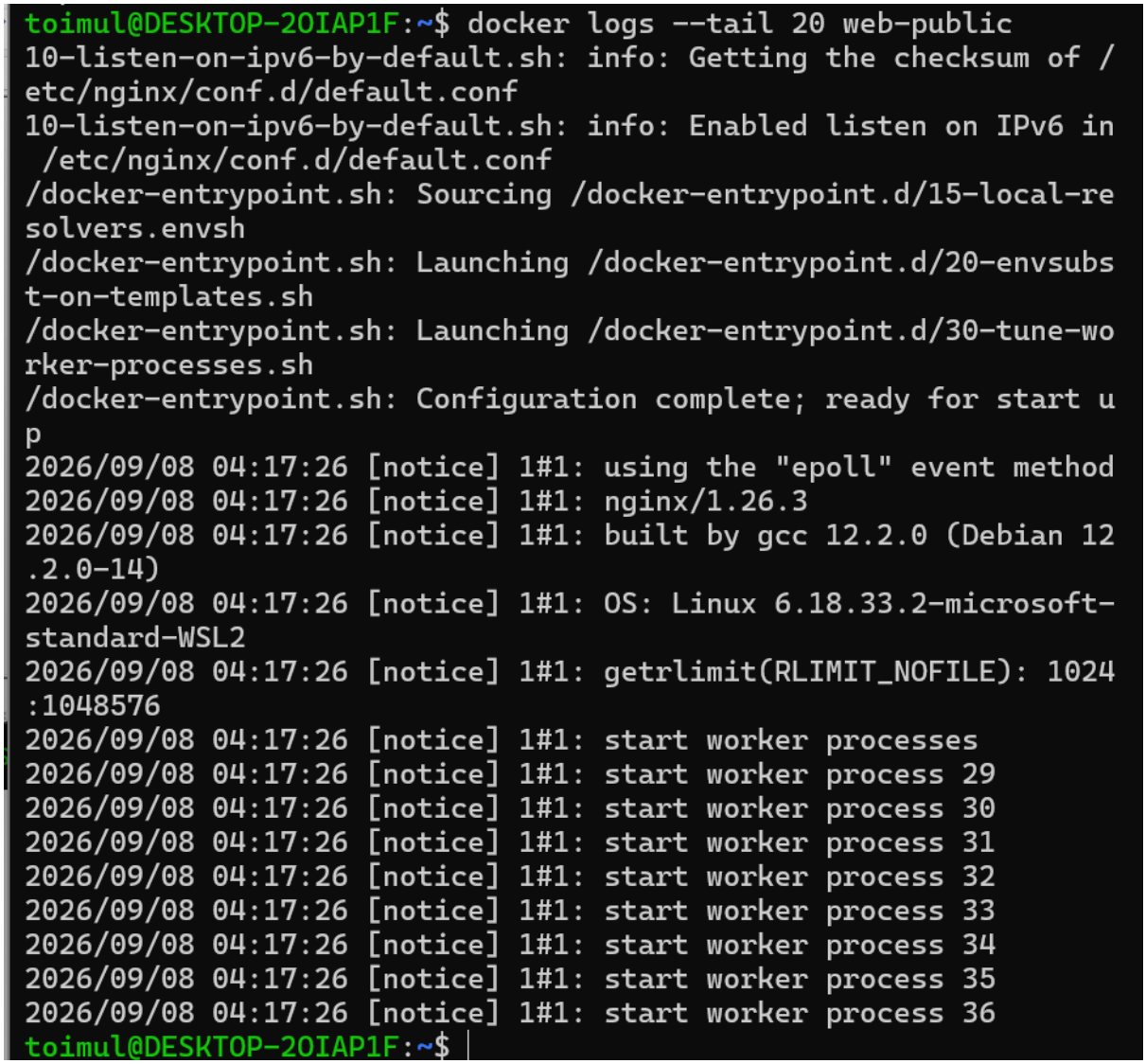
2.1. Penarikan Image Nginx



2.2. Menjalankan Container Web Public



2.3. Inspeksi Log Container



2.4. Uji Akses Service

```
toimul@DESKTOP-2OIAP1F:~$ curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully in
stalled and
working. Further configuration is required.</p>

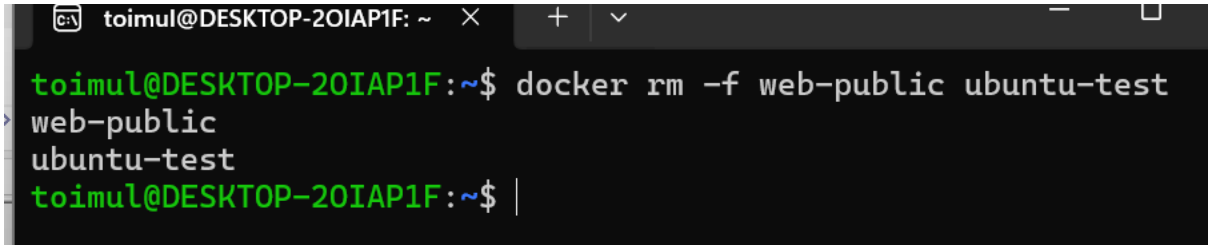
<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

2.5. Eksplorasi Container Ubuntu Interaktif

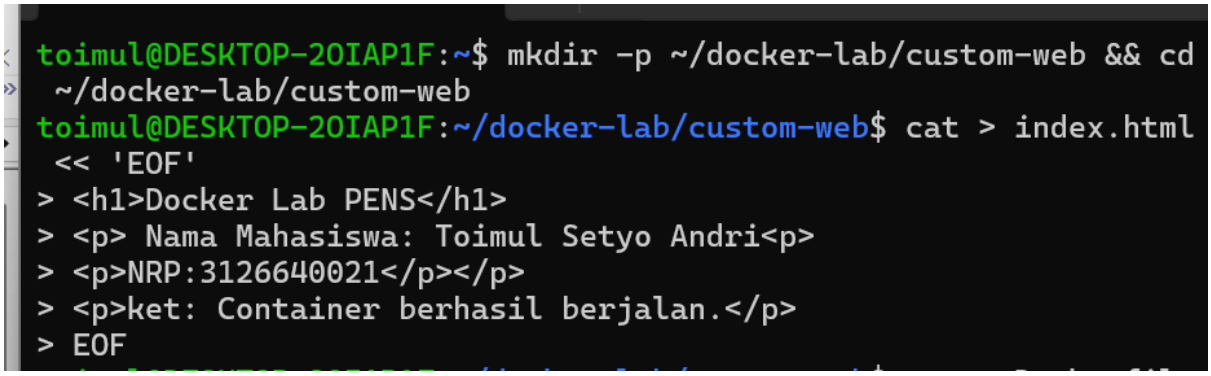
```
toimul@DESKTOP-2OIAP1F:~$ docker run -it --name ubuntu-test ubun
tu:22.04 /bin/bash
Unable to find image 'ubuntu:22.04' locally
22.04: Pulling from library/ubuntu
d544298cabd5: Pull complete
5370ab2c86d9: Download complete
Digest: sha256:2edbbc5dc405e9612ba3584ce95480277e3eb374407b5505f
e26f17df77c7dbc
Status: Downloaded newer image for ubuntu:22.04
root@f26f0c37faff:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-polic
ies/privacy-policy"
UBUNTU_CODENAME=jammy
root@f26f0c37faff:/# exit
exit
```

2.6. Pembersihan Resource

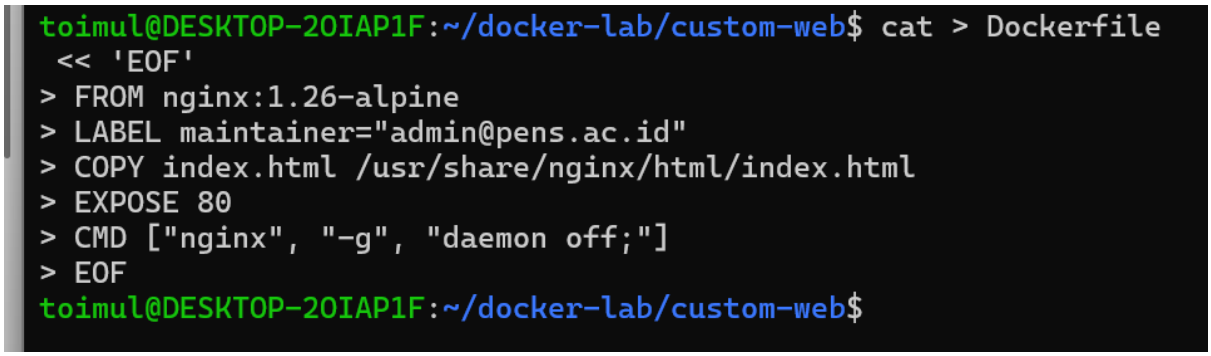


3. Dockerfile custom web statis

3.1. Persiapan Workspace & Web Content



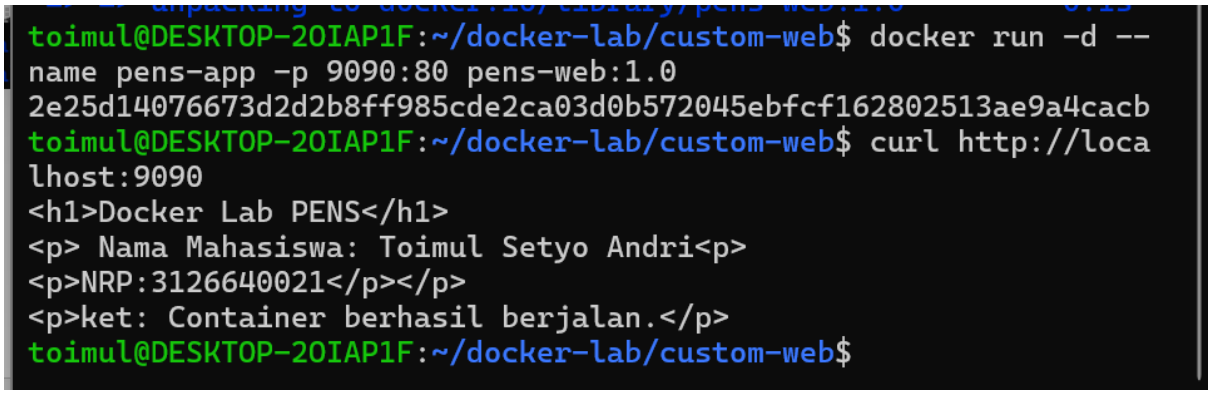
3.2. Penulisan Dockerfile



3.3. Proses Build Image

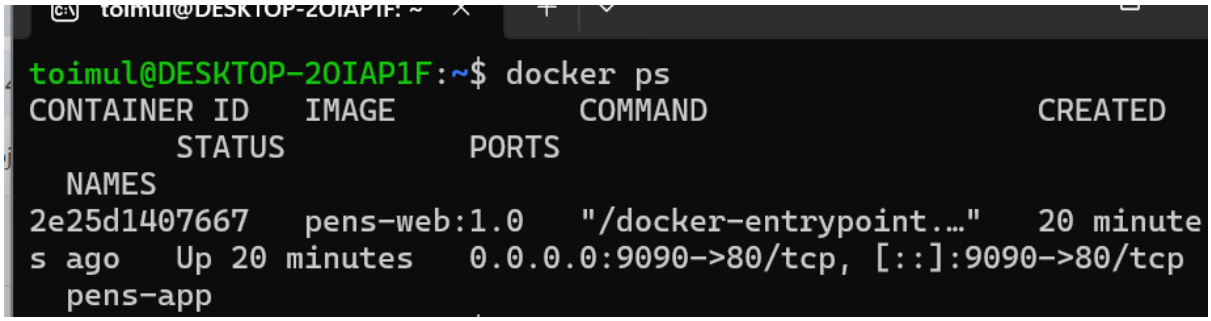


3.4. Deployment & Uji Coba Image Custom

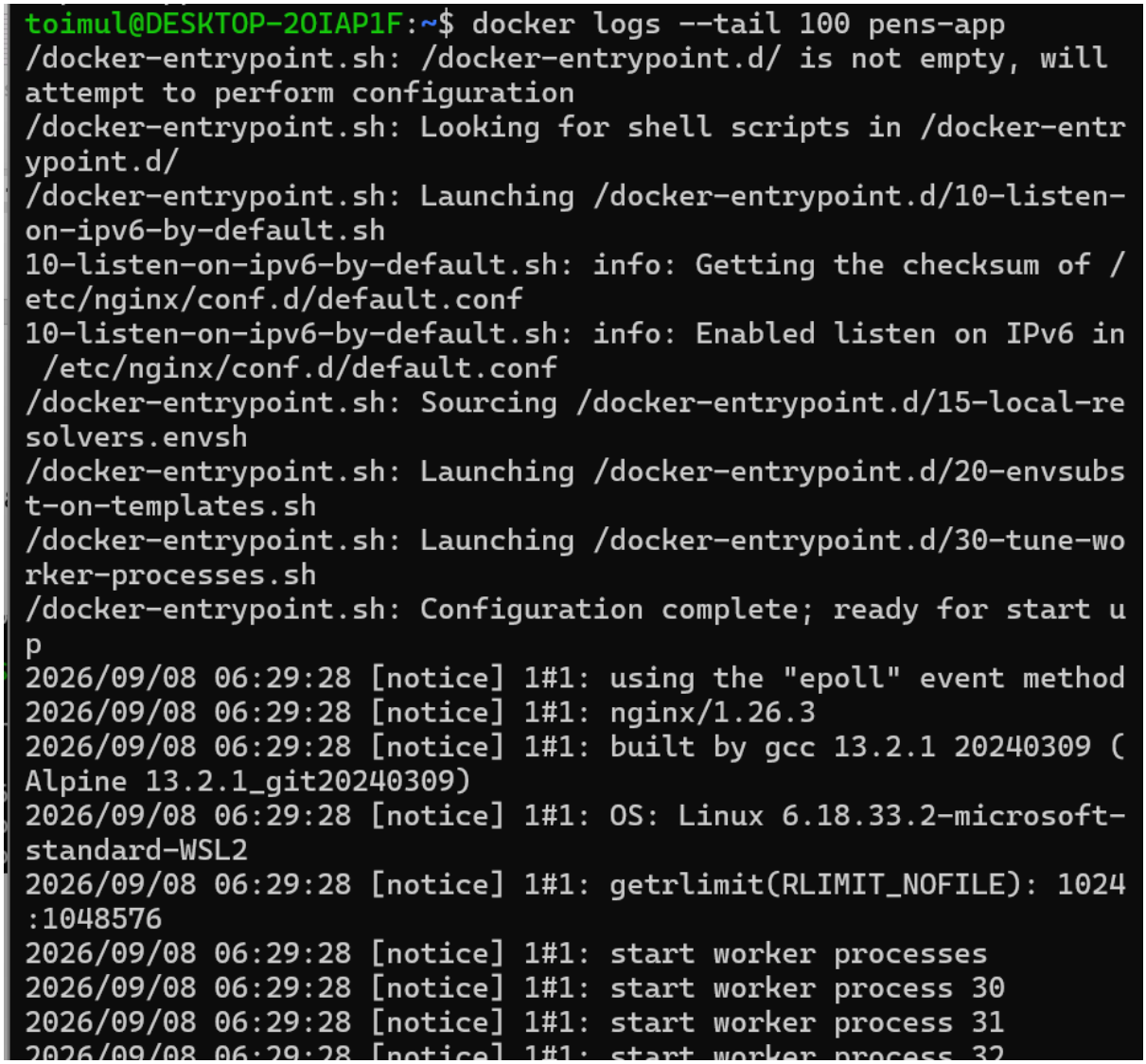


4. Troubleshooting & analisis hasil

4.1 Pemeriksaan Container Berjalan (`docker ps`): Terlihat hanya container `pens-app` yang berjalan pada port `0.0.0.0:9090->80/tcp`.



4.2 Pemeriksaan Log Application (`docker logs --tail 100 pens-app`): Log Nginx menunjukkan status **ready for start up** dan worker process berjalan normal tanpa error.



4.3 Analisis Kegagalan Koneksi (`curl -v http://localhost:8080`):

Gejala: Terjadi error `curl: (7) Failed to connect to localhost port 8080: Connection refused`.

Diagnosis: Kegagalan ini bukan disebabkan oleh kerusakan Docker Engine, melainkan karena container `web-public` yang sebelumnya menggunakan port 8080 telah dihapus pada langkah 2.6 (`docker rm -f web-public ubuntu-test`). Service yang aktif saat ini berada

di port 9090 (`pens-app`).

```
toimul@DESKTOP-2OIAP1F:~$ curl -v http://localhost:8080
* Host localhost:8080 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
*   Trying [::1]:8080...
* connect to ::1 port 8080 from ::1 port 52282 failed: Connectio
n refused
*   Trying 127.0.0.1:8080...
* connect to 127.0.0.1 port 8080 from 127.0.0.1 port 43668 faile
d: Connection refused
* Failed to connect to localhost port 8080 after 3 ms: Could not
 connect to server
* closing connection #0
curl: (7) Failed to connect to localhost port 8080 after 3 ms: C
ould not connect to server
toimul@DESKTOP-2OIAP1F:~$
```

4.4 Inspeksi Network & Runtime (`docker network ls`, `docker inspect`): Driver jaringan berada pada mode `bridge` standar dan status `pens-app` menunjukkan `"Status": running"`, "ExitCode": 0`.

```
toimul@DESKTOP-2OIAP1F:~$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
2f8677fe12df   bridge    bridge    local
bb8d9f57f3c6   host      host      local
57c38be179d8   none      null      local
toimul@DESKTOP-2OIAP1F:~$ docker volume ls
DRIVER    VOLUME NAME
toimul@DESKTOP-2OIAP1F:~$ docker inspect pens-app
[
    {
        "Id": "2e25d14076673d2d2b8ff985cde2ca03d0b572045ebfcf162
802513ae9a4cacb",
        "Created": "2026-09-08T06:29:27.248951855Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
```

Analisis Wajib

1. Analisis Wajib

Masalah yang Muncul: Permintaan HTTP ke http://localhost:8080 mengalami kegagalan koneksi dengan pesan kesalahan curl: (7) Failed to connect to localhost port 8080

Diagnosis:

a) Memeriksa daftar container aktif dengan docker ps, ditemukan bahwa container web-public telah dihapus pada langkah pembersihan *docker rm -f* .
b) Container yang sedang berjalan saat ini adalah pens-app yang dipetakan ke port 9090

Kegagalan koneksi ke port 8080 terjadi karena tidak ada service/container yang mendengarkan pada port tersebut, sehingga permintaan harus diarahkan ke port 9090.

2. Resiko keamanan dan operasional

Resiko keamanan:

Menambahkan user toimul ke grup docker, memberikan akses penuh ke socket Docker Daemon. User non-root tersebut dapat mengeksekusi container dengan bind mount ke sistem file host dan memperoleh hak akses root penuh pada sistem host.

Risiko Operasional:

Menjalankan container dengan perintah docker run manual tanpa batas memori/CPU (resource limits) berpotensi menyebabkan satu container mengonsumsi seluruh sumber daya host (noisy neighbor problem).

3. Rekomendasi Perbaikan

a) **Implementasi Rootless Docker Mode**: Menjalankan Docker Daemon dalam mode *rootless* untuk mencegah eskalasi akses ke host jika terjadi *container breakout*.

b) **Konfigurasi Non-Root User pada Dockerfile**: Mengubah Dockerfile agar tidak menggunakan user root saat menjalankan Nginx, dengan menambahkan direktif USER nginx atau *unprivileged user*.

c) **Penerapan Tagging & SHA-256 Digest**: Mengikat image dasar dengan *digest SHA-256* alih-alih hanya nama tag versi untuk menjamin imutabilitas dan *reproducibility* build.